

Q No.1

Expand $f(z) = \frac{1}{(z-1)^2(z-3)}$ in a Laurent

Series valid for (a) $0 < |z-1| < 2$

and (b) $0 < |z-3| < 2$

Sol:-

(a) $\frac{1}{(z-1)^2(z-3)}$.. $0 < \overset{*}{|z-1|} < 2$

As in part (a) we want only powers of $(z-1)$ and so need to express $z-3$ in terms of $z-1$. This can be done by writing

$$\begin{aligned} f(z) &= \frac{1}{(z-1)^2} \cdot \frac{1}{(z-3)} \\ &= \frac{1}{(z-1)^2} \left[\frac{1}{-2+(z-1)} \right] \Rightarrow \frac{1}{(z-1)^2} \cdot \frac{1}{-2} \left[\frac{1}{1+\left(\frac{z-1}{-2}\right)} \right] \\ &\Rightarrow \frac{1}{(z-1)^2} \cdot \left(-\frac{1}{2}\right) \left[\frac{1}{1-\left(\frac{z-1}{2}\right)} \right] \\ &\Rightarrow -\frac{1}{2(z-1)^2} \left[\frac{1}{1-\left(\frac{z-1}{2}\right)} \right] \end{aligned}$$

$$\Rightarrow -\frac{1}{2(z-1)^2} \left[\frac{1}{1 - \left(\frac{z-1}{2}\right)} \right]$$

1st is valid for
 $|z-1| > 0$
 and ** is
 valid for

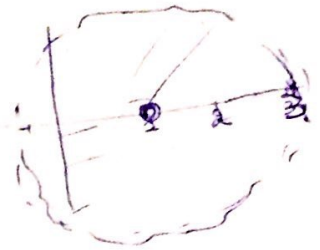
$$\Rightarrow f(z) = -\frac{1}{2(z-1)^2} \left[1 + \left(\frac{z-1}{2}\right) + \right.$$

$$\left| \frac{z-1}{2} \right| < 1$$

$$|z-1| < 2$$

$$\left. \frac{(z-1)^2}{2^2} + \frac{(z-1)^3}{2^3} \dots \right] \text{ so } 0 < |z-1| < 2$$

annular
Region.



$$\Rightarrow -\frac{1}{2(z-1)^2} + \frac{(z-1)}{4(z-1)^2} - \frac{(z-1)^2}{8(z-1)^2} \dots$$

$$\Rightarrow -\frac{1}{2(z-1)^2} - \frac{1}{4(z-1)} - \frac{1}{8} \dots$$

(b) $0 < |z-3| < 2$.

Sol:- In part (b) we want only powers of $z-3$ and so we need to express $(z-1)$ in terms of $z-3$

So

$$f(z) = \frac{1}{(z-1)^2(z-3)} = \frac{1}{(z-3)} \cdot [2 + (z-3)]^{-2}$$

$$\Rightarrow \frac{1}{(z-3)} \cdot \left[\frac{1}{(2 + (z-3))^2} \right]$$

$$\Rightarrow \frac{1}{(z-3)} \cdot \frac{1}{2^2} \left[\frac{1}{\left(1 + \frac{(z-3)}{2}\right)^2} \right]$$

$$\Rightarrow \frac{1}{4(z-3)} \left[\frac{1}{\left(1 + \frac{(z-3)}{2}\right)^2} \right]$$

By using general Binomial theorem:-

$$f(z) = \frac{1}{4(z-3)} \left[1 + \frac{(-2)}{1!} \left(\frac{z-3}{2}\right) + \frac{(-2)(-3)}{2!} \left(\frac{z-3}{2}\right)^2 + \frac{(-2)(-3)(-4)}{3!} \left(\frac{z-3}{2}\right)^3 + \dots \right]$$

(**) Binomial series is valid for $\left|\frac{z-3}{2}\right| < 1$
or $|z-3| < 2$. Multiplying this series by

(*) $\frac{1}{4(z-3)}$ gives Laurent series valid for

$$0 < |z-3| < 2:$$

$$\text{So } f(z) = \frac{1}{4(z-3)} - \frac{1}{4} + \frac{3}{16}(z-3) - \dots$$